

Investigating Data Work Across Domains

New Perspectives on the Work of Creating Data

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ABSTRACT

In the wake of the hype around big data, artificial intelligence, and “data-drivenness,” much attention has been paid to developing novel tools to capitalize upon the deluge of data being recorded and gathered automatically through IT systems. While much of this literature tends to overlook the data itself—sometimes even characterizing it as “data exhaust” that is readily available to be fed into algorithms, which will unlock the insights held within it—a growing body of literature has recently been directed at the (often intensive and skillful) work that goes into creating, collecting, managing, curating, analyzing, interpreting, and communicating data. These investigations detail the practices and processes involved in making data useful and meaningful so that aims of becoming ‘data-driven’ or ‘data-informed’ can become real. Further, In some cases, increased demands for data work have led to the formation of new occupations, whereas at other times data work has been added to the task portfolios of existing occupations and professions, occasionally affecting their core identity. Thus, the evolving forms of data work are requiring individual and organizational resources, new and re-tooled practices and tools, development of new competences and skills, and creation of new functions and roles. While differences exist across the global North and the global

South experience of data work, such factors of data production remain paramount even as they exist largely for the benefit of the data-driven system [21, 32]. This one-day workshop will investigate existing and emerging tasks of data work. Further, participants will seek to understand data work as it impacts: individual data workers; occupations tasked with data work (existing and emerging); organizations (e.g. changing their skill-mix and infrastructuring to support data work); and teaching institutions (grappling with incorporation of data work into educational programs). Participants are required to submit a position paper or a case study drawn from their research to be reviewed and accepted by the organizing committee (submissions should be up to four pages in length). Upon acceptance, participants will read each other’s paper, prepare to shortly present and respond to comments by two discussants and other participants. Subsequently, the workshop will focus on developing a set of core processes and tasks as well as an outline of a research agenda for a CHI-perspective on data work in the coming years.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Human computer interaction (HCI); HCI theory, concepts and models.

KEYWORDS

Data Work, Data-Driven, Datafication, Occupations, Labor

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1 BACKGROUND

1.1 WHY INVESTIGATE DATA WORK?

The creation, maintenance, and use of large scale datasets (a prerequisite for data-driven tools such as data analytics and AI) requires multiple people and forms of human work [15]. While more scholarly attention has been paid to data analytic tools and the content of data, a burgeoning body of research examines this “data work.” Data work is defined as “any human activity related to creating, collecting, managing, curating, analyzing, interpreting, and communicating data” [5]. Data work has profound impacts on both technology design and human labor. A recent study by Sambasivan et al. [32] described the incredibly high prevalence of “data cascades”—negative effects for AI/machine learning tools that are precipitated by undervaluing data quality. Such data quality issues rest directly on data work. Several recent studies also point to the increased labor and burden of engaging with data in a data-driven environment, ranging from physical to emotional burdens [11]. Further, the rise of data work raises crucial questions for organisations.

Thus, studies of data work call attention to the range of socio-technical practices of producing and using data. Importantly, researching data work requires examination of the wide range of formal and informal workers doing labor related to data. While data scientists and data analysts are clearly a growing and important group of data workers [19], many workers are being charged with carrying out data work as part of their jobs (such as teachers and clinicians who are asked to do intensified data entry work). Further, various types of data work are being added to the workloads of clerical and administrative workers who are often undervalued by organizations, which contributes to the invisibility of data work [15]. Data work is also being asked of those who are not formally “workers” at all—for example, implementing patient-reported outcome tools aimed at creating personalized medicine comes with new data work tasks for patients [37].

1.2 Data Work in Different Domains

Data work is not limited to a single sector; rather, it is a growing feature of work across sectors including education, public administration, science, healthcare/public health, management, and more. Below, we briefly describe some of the forces driving data work and a few of the threads of data work research occurring within different sectors.

Science. Recent decades have seen massive interest among scientific communities about how to reap the benefits of the “data deluge” created by expanding IT capabilities [2]. The promise of creating large datasets in machine-readable formats that can be linked and analyzed—and in turn, form the basis for artificial intelligence—has driven huge interest in data intensive science and spawned the creation of the new field of data science. Research has examined the working practices of data scientists. For example, researchers have examined how data scientists work with interpretability tools [19],

and how data scientists collaborate in teams [40]. Research has also examined the interdisciplinary skill sets that must be present in data science teams [7]. Aside from the explicit focus on data science, another key topic has been how scientists share and reuse datasets. This work reveals that data reuse involves collaborative, intensive, and iterative labor (e.g. [30]) including a number of ongoing stewardship practices [18].

Healthcare. Within the healthcare sector, digitization of healthcare has developed from occurring predominantly through small, stand-alone systems for documentation and coordination of healthcare work or clinical databases for research to include (in many settings) large-scale healthcare infrastructures. Most notably, integrated Electronic Patient Records—“EPRs”—also known as Electronic Health Records or “EHRs”) are digital infrastructures that cut across professions, wards, hospitals and regions [1]. Further, the healthcare sector seeks to reach out to the public through initiatives such as ‘patient reported outcome measures’ (PROM) in order to generate data on citizen’s health after or during prolonged treatment. Also, consumer health informatics has provided laypeople and patients with tools to collect, manage and share bodily parameters, whose measurement once required medical equipment and expertise. Thus, the emergence of large-scale information infrastructures has enabled the use of health data for a range of new purposes related to data-driven management, accountability, and performance resource management as well as providing a new source and foundation for healthcare and medical research data [13, 14]. As several authors have pointed out, these processes require new kinds of data work by healthcare occupations and professions, including additional workload to physicians and nurses and the emergence of new occupations such as medical scribes and clinical documentation integrity specialists and new data work tasks being delegated to existing occupations such as birth certificate clerks and medical secretaries [3, 12, 26] and new data work for patients themselves [28].

Education. With the growing datafication in education, schools and classrooms have been naturalized as sites for deploying data tracking, sensing, and analytics technologies [34]. These data-driven technologies have reconfigured educational practices (e.g., teaching, learning, home-school communication, and school management) and relationships among educational actors (i.e. students, teachers, families and parents, schools and districts). These actors have to take on increasing labor in collecting, sensemaking, and communicating both quantifiable data such as test scores and less quantifiable data such as student embodied behavior and emotion. Prior research has shown that the data work undertaken by various educational actors often remains less visible, and manifests broader tensions within the educational system, such as school working environments, unequal access to educational resources, and behaviorist control and surveillance on student bodies [35, 39]. In addition, researchers have also cautioned us that teachers are becoming partly robotized, and teachers’ subjectivities and practices have been increasingly coded and displaced by data and data-driven technologies [39]. The student data collected by teachers are often repurposed by school administrators and parents to evaluate teachers’ performance, which perpetuates the managerial control over teachers [22]. Yet, recent work in CSCW has highlighted that it is undertaking data work that makes space for teachers to negotiate

their discretion and agency in resisting managerial control and enacting care in the classroom [22]. Joining the literature on data work in healthcare, this line of research urges future CSCW/HCI research to better support informal data workers' (such as teachers and caregivers) discretion and resistance in their everyday data work.

Public Administration and Governance. Since around 2013, governments across the globe have undertaken open data programs. However, as critical data studies scholarship has demonstrated, open data does not simply exist, but requires an enormous amount of effort to be created, cleaned, standardized, and managed on new data platforms [8]. Further, government data on its own isn't in itself necessarily useful or meaningful, and frequently needs to be made meaningful through narrativization [9]. The open government movement paved the way for our current moment of data-driven governance and smart cities. Data-driven approaches, particularly after the 2008 global economic crisis, have been pitched as correctives for economic inefficiencies in bureaucracy. This can be seen most clearly in Virginia Eubanks' account of the algorithmic overhaul of administrative welfare service offices, which produced dire effects for both caseworkers and their care, and for welfare recipients, especially along race, class, and ability lines [10]. Further, these modes of governance often beget further investments or additional resources – first in the form of software, hardware, training, and increased data personnel, and later in the form of municipal oversight bodies, policy evaluation studies, and public surveys to oversee the potential pitfalls and measure the efficacy and effects of data-driven programming [17]. This line of research also focuses on the challenges that arise from “inflexible algorithms”, arguing that the algorithm itself lacks these useful characteristics of the street-level bureaucrat that bridge the gap between policy and practice. Altogether, the various methods, technologies, and visions of the data-driven movement are reshaping the labor of on the ground administrative work. While little work to date has examined the situated data work taking place in public administration and governance contexts, this is a context that stands to benefit from further examinations of data work.

AI and Emerging Spaces for Data Work. Data work for machine learning involves the collection, curation, classification, labeling, and verification of data used to train and test machine learning models. These forms of data work have a decisive effect on ML systems [32] and can be performed, among others, by internet users when answering ReCAPTCHA tests, crowd workers who label images, or subject matter experts, for instance, in the healthcare field. A large body of research in HCI and adjacent fields has explored the work that goes into designing [33], creating [24], and understanding [36, 39] data in relation to data-driven systems. Focusing on crowdsourcing platforms and specialized data services, researchers [23, 31] have explored the outsourcing of data work for AI, discussing power asymmetries in terms of occupational status, geographical location, labor conditions, and epistemic authority. Closely related are forms of data work that have emerged as context demands it, for example editing closed captions for online teaching during the pandemic, impersonating AI systems [38] in cases where “the human-in-the-loop” and their labor is kept hidden from the public lens, and forms of content moderation that involve the

evaluation of algorithmic outputs and serve to train automated moderation systems at the same time.

1.3 Workshop Focus: Investigating Data Work Across Domains

This workshop builds on a series of CHI/CSCW workshops that are either fully or partly focused on the topic of data work. These past workshops, which were co-organized or attended by co-organizers of the current WS proposal, focused on specific realms of data work, including data science [20, 24, 25], healthcare [4, 27, 29] and AI [6]. Our WS will build upon the conversations on labor, occupations, skills, collaboration, meaning making, trust, craft, accountability, and power that have been developed at these previous workshops.

Often, scholarship on data work is set within specific domains, as has been the case with prior CHI/CSCW workshops. There are also differences in global experiences of data work [32]. A focus on local domains makes sense methodologically, because understanding data work requires integration of data and domain knowledge. However, the study of data work raises multiple key questions for human-centered computing that cross-cut domains. Holten Møller et al. shows the important role of bringing the data worker herself into the process of data cleaning, contestation and correction to shape data science towards democratic principles of participation as large scale dataset are used for shaping the future workplace [16]. Drawing on the rich contributions of the growing body of literature on data work, the time is now ripe for comparing and framing data work across domains. This workshop will initiate such comparisons in order to identify core themes and issues around data work and to lay out a research framework for investigating data work. Key questions raised by the study of data work across domains include (but are not limited to) the following:

- What are the practices/tasks involved in data work?
- How do multiple data workers coordinate to accomplish data work?
- What are the skills entailed by different forms of data work?
- What digital and analog artifacts are used to carry out data work?
- What new (socio)-technical artifacts/infrastructures have emerged from the rise of data work?
- Who are the data workers? What occupations are emerging to carry out data work, and how are existing occupations changing to meet new demands for data work?
- How do we make data work visible? And, what for—why is making data work visible important?
- How does data work intersect with infrastructure work?
- What are the consequences of data work for workers and organizations?
- What opportunities are there to support data work through design?
- How should we account for data work when considering design, implementation, or evaluation of data-driven tools?
- How does, or how could, the study of data work intersect with labor movements and labor activism?
- What are the politics embedded in data work?
- How does data work relate to the end results produced through data-driven tools, such as data analytics?

- What are the new and emerging fields that require data work?
- What can we learn from data work access disciplines?

This workshop will gather researchers either currently investigating or interested in data work in a variety of sectors and domains to discuss the question above. By building a broad understanding of the practices, occupations, tools, concerns, etc. that characterize data work across domains, we will help to shape a toolbox for the study of data work, a broader agenda for future research in data work, and movement towards an integrated framework for the study of data work.

2 ORGANIZERS

Claus Bossen is an Associate Professor in Information Studies, Aarhus University, Denmark. Based on ethnographic fieldwork and drawing on CSCW and Science & Technology Studies, his research focuses on design, implementation and use of healthcare IT. His past work has examined the impact of digitization and datafication on healthcare practice, healthcare workers, and organizations. Presently, he works on 'making data work visible' with the healthcare domain.

Yunan Chen is an Associate Professor of Informatics at the University of California Irvine. She studies data-driven technologies and datafication in the healthcare context. Her past work examined data-tracking in consumer-facing domains such as fertility, stroke rehabilitation, fitness, and food tracking and the impacts of datafication on individual health management and patient-provider interaction.

Leah Horgan is a PhD Candidate in the Informatics Department at the University of California, Irvine and a 2021-22 Computing Innovation (CI) Fellow at Northeastern University working with the Wylie Lab. Their dissertation work examines the social, political, and material consequences of smart cities and data-driven governance with regard to the labor of civic service, the experience of city living, and the definitional contours of the urban.

Alex Jiahong Lu is a PhD student at the University of Michigan. He studies individuals' everyday navigation of and resistance to data-driven surveillance technologies in different sociopolitical contexts, including classroom behavior surveillance and political communication surveillance in China.

Melissa Mazmanian is a Chancellor's Fellow and Professor of Informatics and Organization and Management at the University of California, Irvine. She is interested in the intersection between information and communication technologies, formal power structures, and everyday practices in organizations. She has engaged in qualitative research on the practices of data production and deployment, such as budgeting and healthcare accountability.

Milagros Miceli is a PhD Candidate at the Technische Universität Berlin and a researcher at the Weizenbaum Institute for the Networked Society. Her research interrogates work practices of machine learning data production and focuses on power differentials in industry contexts as well as their effects on datasets and systems outputs.

Naja Holten Møller is an Assistant Professor in the Department of Computer Science, University of Copenhagen, Denmark and founder of the Confronting Data Co-Lab. Her recent work

raises questions ethnographically to understand: How do evolving data-driven technologies transform conventional forms of work? And, how do we balance concerns for human values while utilizing data in the design of technologies that automate tasks and introduce algorithms in decision-making, for example, in relation to public services?

Gina Neff is the Executive Director of the Minderoo Centre for Technology and Democracy at the University of Cambridge. She studies the future of work in data-rich environments. She is the author of *Venture Labor* (MIT Press, 2012) and, with Dawn Nafus, *Self-Tracking* (MIT Press, 2016) and co-author of the forthcoming *Human-Centered Data Science* (MIT Press, February 2022) with Cecilia Aragon, Shion Guha, Marina Kogan, and Michael Muller.

Kathleen Pine (primary contact for the workshop) is an Assistant Professor in the College of Health Solutions at Arizona State University. Her research examines design, implementation, and use of health information technologies, and has researched how increased demands for data work are reconfiguring healthcare practice, and how healthcare organizations are re-organizing to become "data-driven."

Zhaoyuan Su is a PhD student in the Informatics department at the University of California Irvine. Situated in healthcare contexts, his research examines the design and use of data-driven and algorithmic systems, and their impact on children and caregivers in family life and workplaces.

3 PRE-WORKSHOP PLANS

If accepted, we will create a workshop website with details about the workshop, call for participation, and instructions for submissions to the workshop. The organizers will circulate a brief call for participation along with the link to the workshop on social media (Twitter, Facebook), to multiple list serves (e.g. CSCW-Sig; EUSSET; COOP, and PDC), listservs for those studying STS such as those managed by 4S, and listservs for those interested in labor and technology (e.g. the Labortech google groups email list). We will also circulate the call or participation to personal contacts who might be interested in the workshop. Prior to the workshop, we will make position papers available to participants and ask participants to create a brief (5 minute) presentation with highlights from their position paper.

4 WORKSHOP FORMAT AND STRUCTURE

The workshop will be virtual. We will utilize a video conferencing platform (such as Zoom) to gather participants in a virtual space to carry out all activities planned in the workshop. Instructions for virtual attendance will be distributed to attendees prior to the workshop. Each participant will submit a short position paper (up to 4 pages) presenting a brief overview of their work, along with a specific empirical case from their research. The day will start with participants introducing themselves. Following introductions, accepted papers will be shortly presented by the author(s) to refresh memory. Papers will be discussed by two discussants and through comments from other participants.

After presentations, we will carry out an exercise in which participants identify key themes of the papers, using the list of key

questions above as a starting point. Once the themes are developed, participants will group their research topics into different categories focused on their cases or questions. The workshop will then divide into smaller groups (3-4 persons in break out rooms), which will develop themes, central concepts, and key questions across their individual papers having in mind the overall context of the workshop's papers. Groups will report back in plenum and results be discussed by all.

Groups will then break out again and do a new round of developing themes, central concepts and key questions, but this time across all papers. This will be presented in plenum, followed by an extensive discussion through which a framework for investigating data work across domains will emerge. The workshop will conclude with an ideation session focused on developing a research agenda on data work based on what has been learnt by studying data work across domains.

5 POST-WORKSHOP PLANS

After the workshop, we plan to work with participants to propose a special issue of a journal (e.g. TOCHI) on the theme “investigating data work” (or similar). Co-editors for the special issue will be workshop organizers and participants who are interested in taking on an editorial role. The special issue will allow workshop participants to further develop position papers for publication after the workshop, or to submit other relevant work. The special issue call will be disseminated to non-workshop participants as well to solicit a broad pool of high quality submissions. The introduction of the special issue will review current work on data work and further define a research agenda for data work. We wish to thank Tawanna Dillahunt for her very helpful comments and feedback on this proposal.

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APPENDIX

A CALL FOR PARTICIPATION: “INVESTIGATING DATA WORK ACROSS DOMAINS: NEW PERSPECTIVES ON THE WORK OF CREATING DATA”

Workshop at CHI 2022, DAYOFWEEK MONTH, DATE in
New Orleans, Louisiana, USA and Online

A growing body of literature examines the work that goes into creating, collecting, managing, curating, analyzing, interpreting, and communicating data. These investigations detail the practices and processes involved in making data useful and meaningful so that aims of becoming ‘data-driven’ or ‘data-informed’ can become real. Drawing on these rich contributions, the time is now ripe for comparing and framing data work across domains, occupations, and locales. This workshop will initiate such comparisons in order to identify core themes and issues around data work and to lay out a research framework for investigating it. By building a broad understanding of the practices, occupations, tools, and concerns that characterize data work across domains, we will help to shape a toolbox for the study of data work, a broader agenda for future research in data work, and movement towards an integrated framework for the study of data work.

To apply, submit a position paper or case study from your research of up to 4 pages in length in the ACM short template. Papers should be submitted to Katie Pine (khpine@asu.edu) with the subject header “Data Work Workshop Submission [author(s) last names].” Papers will be reviewed by workshop co-organizers and participants will be selected based on relevance to the workshop theme on the basis of the position papers. Accepted papers will be made public via the workshop website only with permission of the authors (this is optional).

Format: The workshop will be held virtually via a videoconferencing platform such as Zoom. Instructions for virtual attendance will be sent in advance of the workshop.

Important dates:

Submission Deadline: February 15th

Notification: March 7th

Contact: Katie Pine khpine@asu.edu

Workshop Website: TBD